Exhibit index

5 charts, one per patent and product. Claim language is copied from Google Patents. Every quote, source and date was checked against a fresh copy of its page on Nov. 3, 2026; each exhibit ends with its verification log.

EXHIBIT	PATENT	CHARTED AGAINST	STATUS	CLAIM 1 COVERAGE	QUOTES ON PAGE	PRIOR ART	CHECKS FLAGGED
A-1	U.S. 10,469,966 B2 *Zone scene management*	Sonos S2 Controller app (zone scenes) Sonos	Active expires 2027-09-11	●◐◐◐◐◐◐◐ 1 documented · 7 partial · 0 not found	8 of 8	5 refs	7 of 71
A-2	U.S. 8,046,721 B2 *Unlocking a device by performing gestures on an unlock image*	iPhone slide to unlock (iOS 1 to 9) Apple	Expired - Lifetime expired 2025-12-23	●●◐◐ 2 documented · 2 partial · 0 not found	4 of 4	2 refs	2 of 35
A-3	U.S. 5,960,411 A *Method and system for placing a purchase order via a communications network*	Amazon.com 1-Click and Buy Now ordering Amazon	Expired - Lifetime expired 2017-09-12	●●●◐◐◐◐◐ 3 documented · 5 partial · 0 not found	3 of 8	4 refs	11 of 67
A-4	U.S. 7,479,949 B2 *Touch screen device, method, and graphical user interface for determining commands by applying heuristics*	iPhone multi-touch scrolling and swipe gestures Apple	Active expires 2027-09-05	●●●●●●●●◐◐◐◐ 8 documented · 4 partial · 0 not found	7 of 12	3 refs	13 of 87
A-5	U.S. 6,285,999 B1 *Method for node ranking in a linked database*	Google Search ranking by links (PageRank) Google	Expired - Lifetime expired 2018-01-09	●●●● 4 documented · 0 partial · 0 not found	3 of 4	3 refs	4 of 39

U.S. Patent No. 10,469,966 B2

Zone scene management

CHARTED AGAINST Sonos S2 Controller app (zone scenes), Sonos

Sonos lists this patent for the S2 Controller on its patent marking page.

PATENT RECORD — EXHIBIT A-1

ASSIGNEE	Sonos Inc
PRIORITY	Sept. 12, 2006
ISSUED	Nov. 5, 2019
EXPIRES	Sept. 11, 2027
STATUS	Active

From patents.google.com, copy of Nov. 3, 2026. 3 of 3 record checks passed.

8	● 1	◐ 7	○ 0	8/8	3	64/71	Nov. 3, 2026
CLAIM ELEMENTS	DOCUMENTED	PARTIAL	NOT FOUND	QUOTES VERBATIM ON PAGE	SOURCES QUOTED	CHECKS PASSED	RESEARCHED AND CHECKED

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[pre] A computing device comprising: one or more processors;	◐ PARTIAL > "Free Sonos app guides you through setup and provides music control for everyone in the house on all their devices—phones, tablets, and computers."[1] The official guide identifies phones, tablets, and computers as devices running the Sonos controller app. It does not expressly identify the processors in those host devices.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Mar. 2022, during the patent term.
1[a] a non-transitory computer-readable medium; and program instructions stored on the non-transitory computer-readable medium that, when executed by the one or more processors, cause the computing device to perform functions comprising:	◐ PARTIAL > "Download the app from the app store on your mobile device. Open the app and we'll guide you through setup."[2] The S2 guide instructs users to install and run the controller app locally on a mobile device. It does not expressly describe the non-transitory storage medium or execution of stored instructions by processors.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Mar. 2022, during the patent term.

1. *Sonos User Guide*, sonos.com (Mar. 2022), https://www.sonos.com/pdfs/productguides/en-us/sonos-user-guide.pdf (last visited Nov. 3, 2026).
2. *Id.*

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[b] while serving as a controller for a networked media playback system comprising a first zone player and at least two other zone players, wherein the first zone player is operating in a standalone mode in which the first zone player is configured to play back media individually:	◐ PARTIAL > "Saved groups are available on Sonos systems that contain three or more products. Sonos systems with one or two products will not see an option for saved groups."[3] Saved-group functionality requires three or more Sonos products, and the user guide separately describes independent room playback. These sources do not establish the particular three-zone-player configuration or the first player's standalone state during both preset-creation requests.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 1 more supporting source in Appendix B.
1[c] receiving a first request to create a first zone scene comprising a first predefined grouping of zone players including at least the first zone player and a second zone player that are to be configured for synchronous playback of media when the first zone scene is invoked;	● DOCUMENTED > "Select the rooms you want to include in your saved group."[4] The saved-group creation workflow accepts the user's chosen rooms and a group name, then saves that selection as a preset. Sonos's S2 announcement explains that selecting the saved group automatically groups its players, and its grouping documentation specifies synchronized playback.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 2 more supporting sources in Appendix B.
1[d] based on the first request, i) causing creation of the first zone scene, ii) causing an indication of the first zone scene to be transmitted to the first zone player, and iii) causing storage of the first zone scene;	◐ PARTIAL > "If you frequently add the same Sonos rooms together in a group, you can save that group as a preset in the Sonos app. This saved group will show as an option along the top of the Grouping menu."[5] The app creates and retains a room-group preset for later selection. The source does not describe transmitting an indication of that saved preset to the first zone player or identify where the preset is stored.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 1 more supporting source in Appendix B.
1[e] receiving a second request to create a second zone scene comprising a second predefined grouping of zone players including at least the first zone player and a third zone player that are to be configured for synchronous playback of media when the second zone scene is invoked, wherein the third zone player is different than the second zone player;	◐ PARTIAL > "You can create, edit, and delete your saved groups using the Sonos app for iOS or Android."[6] The app supports multiple saved groups and lets users choose each group's rooms. The public instructions do not explicitly demonstrate two presets sharing the first player while containing different second and third players.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page

3. *Saved groups | Sonos*, support.sonos.com, `https://support.sonos.com/en/article/saved-groups` (last visited Nov. 3, 2026).
4. *Id.*
5. *Id.*
6. *Id.*

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[f] based on the second request, i) causing creation of the second zone scene, ii) causing an indication of the second zone scene to be transmitted to the first zone player, and iii) causing storage of the second zone scene; displaying a representation of the first zone scene and a representation of the second zone scene; and while displaying the representation of the first zone scene and the representation of the second zone scene, receiving a third request to invoke the first zone scene; and	PARTIAL > "The names of your saved groups appear above your list of rooms."[7] The documented workflow saves multiple presets, displays their names together, and lets the user select a saved group and Apply it. It does not describe sending the second preset's indication to the first player or expressly demonstrate the two overlapping presets specified in the element.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page
1[g] based on the third request, causing the first zone player to transition from operating in the standalone mode to operating in accordance with the first predefined grouping of zone players such that the first zone player is configured to coordinate with at least the second zone player to output media in synchrony with output of media by at least the second zone player.	PARTIAL > "The new S2 app can save groups of rooms to create groups quickly and easily. Once a group is created, you can select it to start playing music and the players will automatically group up."[8] Selecting a saved S2 group automatically groups its players, and Sonos separately documents that grouped products play the same content in sync. The sources do not explicitly trace the identified first player from standalone operation into that preset or describe its coordination role.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 8, 2020, during the patent term. 1 more supporting source in Appendix B.

7. *Id.*
8. *Introducing S2, New App and OS for Sonos | Sonos Community*, en.community.sonos.com (June 8, 2020), https://en.community.sonos.com/product-updates/introducing-s2-new-app-and-os-for-sonos-6841762 (last visited Nov. 3, 2026).

Appendix A · Prior art

Non-patent literature published before the priority date of Sept. 12, 2006. Dates are checked in Python; anything dated on or after it, or undated, is excluded.

REF.	PUBLISHED	REFERENCE	DISCLOSES	PASSAGE	CHECKS
1	2005-02-11	*NetStreams Offers Additional Features forDigiLinX Multi-Room Audio System* Brian Mitchell[9]	1[c], 1[d], 1[f], 1[g]	"Another highlight of the new firmware upgrade is "Zone-Grouping." This lets installers designate and name particular groups of rooms or listening zones for homeowners to activate at any time."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
2	2005-04	*Sonos Digital Music System User Guide, Version 0504* Sonos[10]	1[a], 1[b], 1[g]	"A zone can be grouped together with any other zone(s) to form a zone group. This will cause all the zones in the zone group to play the same music. You can link or drop zones from a zone group while the music is playing."	● Before priority date ● Non-patent source ● Quote on page · verbatim on an older copy of 2026-06-12; a fresh copy was not available ● Labels match claim
3	2005-02-06	*Review: Sonos Digital Music System* Dan Frakes[11]	1[b], 1[g]	"Using the Zone Menu, you can easily combine zones into linked zones that play the same music in sync."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
4	2005-10-21	*Sonos Digital Music System review: Sonos Digital Music System* Chris Stevens[12]	1[b], 1[g]	"Alternatively, the system can be instructed to play the same track on all your Zoneplayers, and in perfect synchronisation."	● Before priority date ● Non-patent source ● Quote on page · verbatim on an older copy of 2026-01-09; a fresh copy was not available ● Labels match claim
5	2004-09-10	*NetStreams Releases Flagship IP-Based Audio Distribution System* Brian Mitchell[13]	1[pre], 1[a]	"Additionally, because all of the processing code is stored in flash memory, updates to allow DigiLinX to support other new digital audio formats will be easily available in the future."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim

9. Brian Mitchell, *NetStreams Offers Additional Features forDigiLinX Multi-Room Audio System* (Feb. 11, 2005), https://www.ecoustics.com/products/netstreams-offers-additional-features-fordigilinx (last visited Nov. 3, 2026).
10. Sonos, *Sonos Digital Music System User Guide, Version 0504* (Apr. 2005), https://fcc.report/FCC-ID/SBVCR002/664711.pdf (last visited June 12, 2026).
11. Dan Frakes, *Review: Sonos Digital Music System* (Feb. 6, 2005), https://www.macworld.com/article/174329/sonos-4.html (last visited Nov. 3, 2026).
12. Chris Stevens, *Sonos Digital Music System review: Sonos Digital Music System* (Oct. 21, 2005), https://www.cnet.com/reviews/sonos-digital-music-system-review (last visited Jan. 9, 2026).
13. Brian Mitchell, *NetStreams Releases Flagship IP-Based Audio Distribution System* (Sept. 10, 2004), https://www.ecoustics.com/products/netstreams-flagship-ip-based-audio-distribution (last visited Nov. 3, 2026).

Appendix B · Research basis

For each element: Parallel's reasoning, its confidence, and every source it cited. The quoted source is marked ▸.

1[pre] PARTIAL · CONFIDENCE MEDIUM

The guide directly identifies phones, tablets, and computers as controller-app hosts and carries a March 2022 date. The cited passage does not specify a host processor, so the assessment is partial rather than treating ordinary computing-device architecture as expressly documented.

▸ sonos-user-guide · www.sonos.com/pdfs/productguides/en-us/sonos-user-guide.pdf

1[a] PARTIAL · CONFIDENCE MEDIUM

Installation, downloading, opening, and using the S2 app to control the system establish locally running controller software. The passages do not expressly identify a non-transitory medium or processor-execution implementation; those details are therefore left as gaps.

▸ sonos-user-guide · www.sonos.com/pdfs/productguides/en-us/sonos-user-guide.pdf

1[b] PARTIAL · CONFIDENCE MEDIUM

Support requires at least three products for saved groups, the guide describes independent listening, and developer documentation recognizes single-player groups. These are related capabilities, not an explicit account of the specified three-player configuration with the first player remaining standalone during preset creation.

▸ Saved groups | Sonos · support.sonos.com/en/article/saved-groups
sonos-user-guide · www.sonos.com/pdfs/productguides/en-us/sonos-user-guide.pdf

1[c] DOCUMENTED · CONFIDENCE MEDIUM

The creation instructions expressly accept selected rooms and a name for a saved group. The S2 announcement explains invocation and automatic grouping, and grouping support states that grouped products play in sync; together these document the user-facing predefined-group creation behavior.

▸ Saved groups | Sonos · support.sonos.com/en/article/saved-groups
Introducing S2, New App and OS for Sonos | Sonos Community · en.community.sonos.com/product-updates/introducing-s2-new-app-and-os-…
Group and ungroup rooms | Sonos · support.sonos.com/en/article/group-and-ungroup-rooms

1[d] PARTIAL · CONFIDENCE MEDIUM

Saving a selected room group and making it available for later selection supports creation and retention of the preset. None of these passages specifies transmission of the preset's indication to the first player or the storage location, so the compound element is only partially documented.

▸ Saved groups | Sonos · support.sonos.com/en/article/saved-groups
sonos-user-guide · www.sonos.com/pdfs/productguides/en-us/sonos-user-guide.pdf

1[e] PARTIAL · CONFIDENCE MEDIUM

Plural saved groups and room-selection instructions support creating multiple presets with selected membership. The sources do not expressly show the required first-plus-second and first-plus-third overlapping memberships, which is the stated gap.

▸ Saved groups | Sonos · support.sonos.com/en/article/saved-groups

1[f] PARTIAL · CONFIDENCE MEDIUM

The support page describes retained presets, displays saved-group names above the room list, and directs users to select a saved name and Apply. It does not describe the second preset being transmitted to the first player or expressly show the specified overlapping pair of presets.

▸ Saved groups | Sonos · support.sonos.com/en/article/saved-groups

1[g] PARTIAL · CONFIDENCE MEDIUM

The June 8, 2020 S2 announcement directly states that selecting a saved group automatically groups its players. Grouping support establishes synchronized content, but neither passage expressly traces the specified first player's standalone-to-group transition or identifies its coordination role.

▸ Introducing S2, New App and OS for Sonos | Sonos Community · en.community.sonos.com/product-updates/introducing-s2-new-app-and-os-…
Group and ungroup rooms | Sonos · support.sonos.com/en/article/group-and-ungroup-rooms

Appendix C · Verification log

Every check run on this exhibit. Green passed, amber needs a person to look, red failed; details for every flag follow the grid.

CLAIM LANGUAGE	EVIDENCE	PRIOR ART	QUOTES, SOURCES AND DATES	PREPARED WITH
Extract API on patents.google.com, copy of Nov. 3, 2026. Split into elements in Python; no model writes claim text.	Task API, pro processor, patent databases excluded. `srun_728f2b03f51547d34170747ea83f9414`	Task API, pro processor, dates enforced in Python. `srun_728f2b03f51547d34d05f738a0558ea1`	Extract API re-reads every cited page from a copy at most a day old; Python compares each quote word for word and checks domains and dates.	parallel

PATENT RECORD	PATENT NUMBER	CLAIM 1 VERBATIM	DATES IN ORDER
US10469966B2	●	●	●

CLAIM ELEMENT	RESEARCHED	QUOTE AND SOURCE	QUOTE ON PAGE	NON-PATENT SOURCE	IN RESEARCH BASIS	DATED
1[pre]	●	●	●	●	●	●
1[a]	●	●	●	●	●	●
1[b]	●	●	●	●	●	●
1[c]	●	●	●	●	●	●
1[d]	●	●	●	●	●	●
1[e]	●	●	●	●	●	●
1[f]	●	●	●	●	●	●
1[g]	●	●	●	●	●	●

PRIOR ART	BEFORE PRIORITY DATE	NON-PATENT SOURCE	QUOTE ON PAGE	LABELS MATCH CLAIM
Ref. 1	●	●	●	●
Ref. 2	●	●	●	●
Ref. 3	●	●	●	●
Ref. 4	●	●	●	●
Ref. 5	●	●	●	●

FLAGGED FOR REVIEW · 7

- 1[b] · Source dated · no date on the page
- 1[c] · Source dated · no date on the page
- 1[d] · Source dated · no date on the page
- 1[e] · Source dated · no date on the page
- 1[f] · Source dated · no date on the page

- **Ref. 2** · Quote found on the page · verbatim on an older copy of 2026-06-12; a fresh copy was not available

- **Ref. 4** · Quote found on the page · verbatim on an older copy of 2026-01-09; a fresh copy was not available

U.S. Patent No. 8,046,721 B2

Unlocking a device by performing gestures on an unlock image

CHARTED AGAINST iPhone slide to unlock (iOS 1 to 9), Apple

Patent owner's own product. Patent expired in 2025.

PATENT RECORD — EXHIBIT A-2

ASSIGNEE	Apple Inc
PRIORITY	Dec. 23, 2005
ISSUED	Oct. 25, 2011
EXPIRED	Dec. 23, 2025
STATUS	Expired - Lifetime

From patents.google.com, copy of Nov. 3, 2026. 3 of 3 record checks passed.

4	● 2	◐ 2	○ 0	4/4	3	33/35	Nov. 3, 2026
CLAIM ELEMENTS	DOCUMENTED	PARTIAL	NOT FOUND	QUOTES VERBATIM ON PAGE	SOURCES QUOTED	CHECKS PASSED	RESEARCHED AND CHECKED

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[pre] A method of unlocking a hand-held electronic device, the device including a touch-sensitive display, the method comprising:	● DOCUMENTED > "To unlock the first iPhone, you touched the arrow button on the lock screen and dragged it over to the right."[1] The original iPhone is unlocked by touching an on-screen arrow and dragging it right, documenting a touch-display unlocking method.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 29, 2017, during the patent term. 1 more supporting source in Appendix B.
1[a] detecting a contact with the touch-sensitive display at a first predefined location corresponding to an unlock image;	● DOCUMENTED > "To unlock the first iPhone, you touched the arrow button on the lock screen and dragged it over to the right."[2] For the original arrow-slider implementation, the user begins by touching the displayed arrow button before dragging it right, supporting an initial graphical touch target. This evidence should not be generalized to every version in iOS 1–9: iOS 7 allowed unlocking swipes from anywhere on the lock screen.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 29, 2017, during the patent term. 1 more supporting source in Appendix B.

1. *The original iPhone in photos: Looking back on the iPhone that started it all*, macworld.com (June 29, 2017), https://www.macworld.com/article/230231/original-2007-iphone-photo-album.html (last visited Nov. 3, 2026).
2. *Id.*

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[b] continuously moving the unlock image on the touch-sensitive display in accordance with movement of the contact while continuous contact with the touch screen is maintained, wherein the unlock image is a graphical, interactive user-interface object with which a user interacts in order to unlock the device; and	◐ PARTIAL > "As soon as you power it up, you're presented with a software button with an arrow on it that sits at the bottom of the screen in a channel with the message 'Slide to Unlock' that animates to suggest that you slide the button to the right."[3] The review describes a graphical arrow button in a channel that users slide right, but does not explicitly describe uninterrupted contact or continuous tracking of the image with the finger. An iOS 7 review separately reports movement of a whole screen layer rather than a small slider, so the earlier mechanism is not established across the full iOS 1–9 range.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 1 more supporting source in Appendix B.
1[c] unlocking the hand-held electronic device if the moving the unlock image on the touch-sensitive display results in movement of the unlock image from the first predefined location to a predefined unlock region on the touch-sensitive display.	◐ PARTIAL > "Either way, the on-screen slider appears, but your iPhone doesn't actually awaken until you drag the slider to the right with your finger."[4] The guide links a rightward finger drag of the on-screen slider to unlocking the iPhone. It does not identify the slider's required destination or establish that reaching a predefined display region is the condition that triggers unlocking.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Oct. 6, 2021, during the patent term.

3. *iPhone 1.0: The 10th Anniversary MacStories Review*, macstories.net, https://www.macstories.net/reviews/iphone-1-0-the-10th-anniversary-macstories-review (last visited Nov. 3, 2026).
4. *How to Lock and Unlock Your iPhone*, dummies.com (Oct. 6, 2021), https://www.dummies.com/article/technology/electronics/cell-phones/iphones/how-to-lock-and-unlock-your-iphone-196930 (last visited Nov. 3, 2026).

Appendix A · Prior art

Non-patent literature published before the priority date of Dec. 23, 2005. Dates are checked in Python; anything dated on or after it, or undated, is excluded.

REF.	PUBLISHED	REFERENCE	DISCLOSES	PASSAGE	CHECKS
1	1990-11	*Touchscreen Toggle Switches: Push or Slide? Design Issues and Usability Study* Catherine Plaisant and Daniel Wallace[5]	1[a], 1[b], 1[c]	"Users can then grab the pointer and slide it to the other side. If the finger is released before reaching the other side the pointer springs back to its previous position."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
2	2004-07-29	*N1 Quick Start Guide, V0.5* Neonode[6]	1[pre], 1[c]	"The text "Right sweep to unlock" appears on the screen. Sweep right to unlock your unit."	● Before priority date · 2004-07-29 vs priority 2005-12-23, read from the note "Not printed; publicly archived on 2004-07-29, before the priority date": check by hand ● Non-patent source ● Quote on page ● Labels match claim

5. Catherine Plaisant and Daniel Wallace, *Touchscreen Toggle Switches: Push or Slide? Design Issues and Usability Study* (Nov. 1990), http://www.cs.umd.edu/hcil/trs/90-08/90-08.pdf (last visited Nov. 3, 2026).
6. Neonode, *N1 Quick Start Guide, V0.5* (July 29, 2004), https://www.courtlistener.com/docket/4178994/2121/15/apple-inc-v-samsung-electronics-co-ltd (last visited Nov. 3, 2026).

Appendix B · Research basis

For each element: Parallel's reasoning, its confidence, and every source it cited. The quoted source is marked ▸.

1[pre] DOCUMENTED · CONFIDENCE MEDIUM

Macworld directly describes touching and dragging an on-screen arrow to unlock the original iPhone, and its title/date segments establish the source metadata. MacStories explicitly identifies the iPhone's touch screen. These are review sources rather than primary technical documentation.

▸ Original iPhone 2007 Photo Album | Macworld · www.macworld.com/article/230231/original-2007-iphone-photo-album.html
iPhone 1.0: The 10th Anniversary MacStories Review - MacStories · www.macstories.net/reviews/iphone-1-0-the-10th-anniversary-macstories…

1[a] DOCUMENTED · CONFIDENCE MEDIUM

The Macworld sentence identifies the displayed arrow as the initial touch target for the original implementation. The iOS 7 description permits swiping anywhere on the lock screen, supporting the explicit limitation on generalizing that fixed-target evidence across the entire requested version range.

▸ Original iPhone 2007 Photo Album | Macworld · www.macworld.com/article/230231/original-2007-iphone-photo-album.html
Get to know iOS 7: Design changes | Macworld · www.macworld.com/article/221891/get-to-know-ios-7-design-changes.html

1[b] PARTIAL · CONFIDENCE MEDIUM

MacStories describes a software arrow button in a channel and direct finger manipulation of screen elements, supporting the graphical interactive-object portion. It does not explicitly describe uninterrupted contact or continuous image tracking. AppleInsider directly reports that iOS 7 moves an entire screen layer instead of the small slider, supporting the version caveat; no publication date was shown for the quoted MacStories page.

▸ iPhone 1.0: The 10th Anniversary MacStories Review - MacStories · www.macstories.net/reviews/iphone-1-0-the-10th-anniversary-macstories…
Inside Apple's iOS 7: Details on the new features of your just-updated iPhone & iPad | AppleInsider · appleinsider.com/articles/13/09/18/inside-apples-ios-7-details-on-the…

1[c] PARTIAL · CONFIDENCE MEDIUM

The guide's unlocking context and quoted instruction connect rightward finger dragging of the slider with unlocking. The cited passage supplies no required endpoint or predefined destination region, so it supports only a partial assessment. The page explicitly supplies the title and update date.

▸ How to Lock and Unlock Your iPhone | dummies · www.dummies.com/article/technology/electronics/cell-phones/iphones/ho…

Appendix C · Verification log

Every check run on this exhibit. Green passed, amber needs a person to look, red failed; details for every flag follow the grid.

CLAIM LANGUAGE	EVIDENCE	PRIOR ART	QUOTES, SOURCES AND DATES	PREPARED WITH
Extract API on patents.google.com, copy of Nov. 3, 2026. Split into elements in Python; no model writes claim text.	Task API, pro processor, patent databases excluded. `srun_728f2b03f51547d3108e5b709ab6bb05`	Task API, pro processor, dates enforced in Python. `srun_728f2b03f51547d3897d9300ef73e67d`	Extract API re-reads every cited page from a copy at most a day old; Python compares each quote word for word and checks domains and dates.	parallel

PATENT RECORD	PATENT NUMBER	CLAIM 1 VERBATIM	DATES IN ORDER
US8046721B2	●	●	●

CLAIM ELEMENT	RESEARCHED	QUOTE AND SOURCE	QUOTE ON PAGE	NON-PATENT SOURCE	IN RESEARCH BASIS	DATED
1[pre]	●	●	●	●	●	●
1[a]	●	●	●	●	●	●
1[b]	●	●	●	●	●	◐
1[c]	●	●	●	●	●	●

PRIOR ART	BEFORE PRIORITY DATE	NON-PATENT SOURCE	QUOTE ON PAGE	LABELS MATCH CLAIM
Ref. 1	●	●	●	●
Ref. 2	◐	●	●	●

FLAGGED FOR REVIEW · 2

- ◐ **1[b]** · Source dated · no date on the page
- ◐ **Ref. 2** · Published before the priority date · 2004-07-29 vs priority 2005-12-23, read from the note “Not printed; publicly archived on 2004-07-29, before the priority date”: check by hand

U.S. Patent No. 5,960,411 A

Method and system for placing a purchase order via a communications network

CHARTED AGAINST Amazon.com 1-Click and Buy Now ordering, Amazon

Patent owner's own product. Patent expired in 2017.

PATENT RECORD

EXHIBIT A-3

ASSIGNEE	Amazon.com Inc
PRIORITY	Sept. 12, 1997
ISSUED	Sept. 28, 1999
EXPIRED	Sept. 12, 2017
STATUS	Expired - Lifetime

From patents.google.com, copy of Nov. 3, 2026. 3 of 3 record checks passed.

8	● 3	◐ 5	○ 0	3/8	5	56/67	Nov. 3, 2026
CLAIM ELEMENTS	DOCUMENTED	PARTIAL	NOT FOUND	QUOTES VERBATIM ON PAGE	SOURCES QUOTED	CHECKS PASSED	RESEARCHED AND CHECKED

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[pre] A method of placing an order for an item comprising:	● DOCUMENTED > "1-Click securely stores billing and shipping information so that each customer needs only one click of the mouse to buy a selected item, rather than entering the same information over and over again for each purchase."[1] Amazon describes 1-Click as a way to buy a selected item with one click using stored billing and shipping information.	CONFIDENCE: HIGH ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Oct. 11, 1999, during the patent term.
1[a] under control of a client system, displaying information identifying the item; and	● DOCUMENTED > "This page will contain information about the specific product and will have an option to 'buy now with 1-Click.'"[2] The court record describes an Amazon product detail page containing information about the product and a 1-Click purchase option. The page does not show a publication date in the evidence reviewed.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · verbatim on an older copy of 2026-02-13; a fresh copy was not available ● Non-patent source ● In research basis ● Dated · no date on the page

1. *Amazon.com Receives Patent for 1-click*, press.aboutamazon.com (Oct. 11, 1999), https://press.aboutamazon.com/1999/10/amazon-com-receives-patent-for-1-click (last visited Nov. 3, 2026).
2. *IPXL Holdings, LLC v. Amazon.Com, Inc., 333 F. Supp. 2d 513 (E.D. Va. 2004)*, law.justia.com, https://law.justia.com/cases/federal/district-courts/FSupp2/333/513/2500769/ (last visited Feb. 13, 2026).

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[b] in response to only a single action being performed, sending a request to order the item along with an identifier of a purchaser of the item to a server system;	◐ PARTIAL > "1-Click securely stores billing and shipping information so that each customer needs only one click of the mouse to buy a selected item, rather than entering the same information over and over again for each purchase."[3] Amazon describes buying a selected item with one click; a court opinion separately says cookies transferred a sessionID in the accused system. The public evidence reviewed does not establish that a purchaser identifier was sent with that specific one-click order request.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Oct. 11, 1999, during the patent term. 1 more supporting source in Appendix B.
1[c] under control of a single-action ordering component of the server system, receiving the request;	◐ PARTIAL > "By choosing the 'buy now with 1-Click' option from this product detail page, the user initiates the product ordering and shipping process."[4] The court record describes the 1-Click option initiating the ordering and shipping process. It does not describe receipt by a particular server-side single-action ordering component.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · verbatim on an older copy of 2026-02-13; a fresh copy was not available ● Non-patent source ● In research basis ● Dated · no date on the page
1[d] retrieving additional information previously stored for the purchaser identified by the identifier in the received request; and	◐ PARTIAL > "Amazon's customers can store payment information and shipping addresses in their Amazon customer accounts. This information can then be retrieved later when that customer uses the 1-Click features."[5] The appellate opinion says customer payment and shipping information is stored and can later be retrieved when the customer uses 1-Click. It does not expressly tie that retrieval to a purchaser identifier in the received request.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · not on the page, copy of 2026-11-03 ● Non-patent source ● In research basis · not among this element's citations ● Dated Dated Nov. 2, 2011, during the patent term. 1 more supporting source in Appendix B.

3. *Amazon.com Receives Patent for 1-click*, *supra* note 1.
4. *IPXL Holdings, LLC v. Amazon.Com, Inc., 333 F. Supp. 2d 513 (E.D. Va. 2004)*, *supra* note 2.
5. *Amazon.com, Inc. v. Barnesandnoble.com, Inc., No. 00-1109 — U.S. Court of Appeals, Federal Circuit Opinion*, storage.courtlistener.com (Nov. 2, 2011), https://storage.courtlistener.com/harvard_pdf/772066.pdf (last visited Nov. 3, 2026).

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[e] generating an order to purchase the requested item for the purchaser identified by the identifier in the received request using the retrieved additional information; and	◐ PARTIAL > "The customer's 1-Click order is placed in a holding state for 90 minutes."[6] The court record describes a 1-Click order being placed and later finalized if not cancelled. The evidence does not specify that the order is generated using information retrieved for a purchaser identified in the request.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · verbatim on an older copy of 2026-02-13; a fresh copy was not available ● Non-patent source ● In research basis ● Dated · no date on the page
1[f] fulfilling the generated order to complete purchase of the item	● DOCUMENTED > "Amazon then checks which items it can group together, calculates shipping costs, and then sends your order out for delivery."[7] The 2017 description of Amazon 1-Click says Amazon sends the order out for delivery after grouping items and calculating shipping costs.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Aug. 2, 2017, during the patent term.
1[g] whereby the item is ordered without using a shopping cart ordering model.	◐ PARTIAL > "Once activated, you can skip the standard checkout process and buy something by clicking a single button."[8] The guide describes 1-Click as skipping standard checkout with a single button. It does not expressly establish that the ordering route avoids a shopping cart model.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · page could not be fetched; check by hand ● Non-patent source ● In research basis ● Dated Dated Aug. 2003, during the patent term.

6. *IPXL Holdings, LLC v. Amazon.Com, Inc., 333 F. Supp. 2d 513 (E.D. Va. 2004)*, *supra* note 2.
7. *What Is 1-Click Ordering On Amazon and How Does It Work?*, howtogeek.com (Aug. 2, 2017), https://www.howtogeek.com/320394/what-is-1-click-ordering-on-amazon-and-how-does-it-work (last visited Nov. 3, 2026).
8. *Enable 1-Click Buying*, oreilly.com (Aug. 2003), https://www.oreilly.com/library/view/amazon-hacks/0596005423/ch02s04.html.

Appendix A · Prior art

Non-patent literature published before the priority date of Sept. 12, 1997. Dates are checked in Python; anything dated on or after it, or undated, is excluded.

REF.	PUBLISHED	REFERENCE	DISCLOSES	PASSAGE	CHECKS
1	1995-08	*NetBill: An Internet Commerce System Optimized for Network-Delivered Services* Marvin Sirbu and J. D. Tygar[9]	1[a], 1[b], 1[c], 1[d], 1[f], 1[g]	"The application can present the price quote to the consumer or it can approve the purchase without prompting the consumer. For example, the consumer may specify that her client software accept any price quote below some threshold amount; this relieves her of the burden of assenting to every low-value price quote via a dialog box."	● Before priority date ● Non-patent source ● Quote on page · page could not be fetched; check by hand ● Labels match claim
2	1995-07	*Payment Switches for Open Networks* David K. Gifford, Lawrence C. Stewart, Andrew C. Payne, and G. Winfield Treese[10]	1[a], 1[f], 1[g]	"The product description page is returned to the client and displayed. The client then activates one of the payment URLs to purchase the product the URL describes."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
3	1995-12-11	*The Millicent Protocol for Inexpensive Electronic Commerce* Steve Glassman, Mark Manasse, Martín Abadi, Paul Gauthier, and Patrick Sobalvarro[11]	1[f], 1[g]	"To make a purchase, the customer sends the request, scrip, and a "signature" of the request to the vendor."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
4	1996-06	*Perils and pitfalls of practical cybercommerce* Nathaniel S. Borenstein et al.[12]	1[d]	"When First Virtual processes a financial transaction, it looks up the buyer's Virtual personal identification number (PIN) in its database and finds the buyer's email address of record."	● Before priority date ● Non-patent source ● Quote on page · not on the page, copy of 2026-11-03 ● Labels match claim

9. Marvin Sirbu and J. D. Tygar, *NetBill: An Internet Commerce System Optimized for Network-Delivered Services* (Aug. 1995), https://kilthub.cmu.edu/ndownloader/files/12236102.
10. David K. Gifford, Lawrence C. Stewart, Andrew C. Payne, and G. Winfield Treese, *Payment Switches for Open Networks* (July 1995), https://usenix.org/legacy/publications/library/proceedings/ec95/full_papers/gifford.txt (last visited Nov. 3, 2026).
11. Steve Glassman, Mark Manasse, Martín Abadi, Paul Gauthier, and Patrick Sobalvarro, *The Millicent Protocol for Inexpensive Electronic Commerce* (Dec. 11, 1995), https://www.w3.org/Conferences/WWW4/Papers/246 (last visited Nov. 3, 2026).
12. Nathaniel S. Borenstein et al., *Perils and pitfalls of practical cybercommerce* (June 1996), https://dl.acm.org/doi/pdf/10.1145/228503.228510 (last visited Nov. 3, 2026).

Appendix B · Research basis

For each element: Parallel's reasoning, its confidence, and every source it cited. The quoted source is marked ▸.

1[pre] DOCUMENTED · CONFIDENCE HIGH

Amazon's 1999 release directly describes purchasing a selected item with one click and using stored billing and shipping information.

▸ Amazon.com-Receives-Patent-for-1-click - US Press Center · press.aboutamazon.com/1999/10/amazon-com-receives-patent-for-1-click

1[a] DOCUMENTED · CONFIDENCE MEDIUM

The court record states that the product detail page contains information about the specific product and a 1-Click option.

▸ IPXL Holdings, LLC v. Amazon. Com, Inc., 333 F. Supp. 2d 513 (E.D. Va. 2004) :: Justia · law.justia.com/cases/federal/district-courts/FSupp2/333/513/2500769/

1[b] PARTIAL · CONFIDENCE MEDIUM

The release documents one-click purchase; the appellate opinion separately states that cookies transferred a sessionID in the accused system. The evidence does not explicitly connect that identifier transfer to the same one-click order request.

▸ Amazon.com-Receives-Patent-for-1-click - US Press Center · press.aboutamazon.com/1999/10/amazon-com-receives-patent-for-1-click
Cordance Corp. v. Amazon.com, Inc., No. 2010-1502 — U.S. Court of Appeals for the Federal Circuit Opinion · storage.courtlistener.com/recap/gov.uscourts.ded.36947.600.1.pdf

1[c] PARTIAL · CONFIDENCE MEDIUM

The court record describes the customer initiating the ordering and shipping process through the 1-Click option, but does not identify a server-side component receiving the request.

▸ IPXL Holdings, LLC v. Amazon. Com, Inc., 333 F. Supp. 2d 513 (E.D. Va. 2004) :: Justia · law.justia.com/cases/federal/district-courts/FSupp2/333/513/2500769/

1[d] PARTIAL · CONFIDENCE MEDIUM

The opinion states that payment details and shipping addresses are stored in customer accounts and can later be retrieved when that customer uses 1-Click; it does not explicitly state that the request identifier selects that customer.

Cordance Corp. v. Amazon.com, Inc., No. 2010-1502 — U.S. Court of Appeals for the Federal Circuit Opinion · storage.courtlistener.com/recap/gov.uscourts.ded.36947.600.1.pdf

1[e] PARTIAL · CONFIDENCE MEDIUM

The court record confirms an order is placed in a holding state, but does not explain order-generation mechanics using information retrieved for an identified purchaser.

▸ IPXL Holdings, LLC v. Amazon. Com, Inc., 333 F. Supp. 2d 513 (E.D. Va. 2004) :: Justia · law.justia.com/cases/federal/district-courts/FSupp2/333/513/2500769/

1[f] DOCUMENTED · CONFIDENCE MEDIUM

The 2017 article states that Amazon sends the order out for delivery after checking grouping and shipping costs.

▸ What Is 1-Click Ordering On Amazon and How Does It Work? · www.howtogeek.com/320394/what-is-1-click-ordering-on-amazon-and-how-d…

1[g] PARTIAL · CONFIDENCE MEDIUM

The 2003 guide says 1-Click skips standard checkout, but does not specifically say it bypasses the shopping cart ordering model.

▸ Enable 1-Click Buying - Amazon Hacks [Book] · www.oreilly.com/library/view/amazon-hacks/0596005423/ch02s04.html

Appendix C · Verification log

Every check run on this exhibit. Green passed, amber needs a person to look, red failed; details for every flag follow the grid.

CLAIM LANGUAGE
Extract API on patents.google.com, copy of Nov. 3, 2026. Split into elements in Python; no model writes claim text.

EVIDENCE
Task API, pro processor, patent databases excluded.
`srun_728f2b03f51547d3dfac42667b3dddba`

PRIOR ART
Task API, pro processor, dates enforced in Python.
`srun_728f2b03f51547d31c23de36dda4f006`

QUOTES, SOURCES AND DATES
Extract API re-reads every cited page from a copy at most a day old; Python compares each quote word for word and checks domains and dates.

PREPARED WITH
parallel

PATENT RECORD	PATENT NUMBER	CLAIM 1 VERBATIM	DATES IN ORDER
US5960411A	●	●	●

CLAIM ELEMENT	RESEARCHED	QUOTE AND SOURCE	QUOTE ON PAGE	NON-PATENT SOURCE	IN RESEARCH BASIS	DATED
1[pre]	●	●	●	●	●	●
1[a]	●	●	●	●	●	●
1[b]	●	●	●	●	●	●
1[c]	●	●	●	●	●	●
1[d]	●	●	●	●	●	●
1[e]	●	●	●	●	●	●
1[f]	●	●	●	●	●	●
1[g]	●	●	●	●	●	●

PRIOR ART	BEFORE PRIORITY DATE	NON-PATENT SOURCE	QUOTE ON PAGE	LABELS MATCH CLAIM
Ref. 1	●	●	●	●
Ref. 2	●	●	●	●
Ref. 3	●	●	●	●
Ref. 4	●	●	●	●

FLAGGED FOR REVIEW · 11

- 1[a] · Quote found on the page · verbatim on an older copy of 2026-02-13; a fresh copy was not available
- 1[a] · Source dated · no date on the page
- 1[c] · Quote found on the page · verbatim on an older copy of 2026-02-13; a fresh copy was not available
- 1[c] · Source dated · no date on the page
- 1[d] · Quote found on the page · not on the page, copy of 2026-11-03
- 1[d] · Source cited in the research basis · not among this element's citations

- `1[e]` · Quote found on the page · verbatim on an older copy of 2026-02-13; a fresh copy was not available
- `1[e]` · Source dated · no date on the page
- `1[g]` · Quote found on the page · page could not be fetched; check by hand
- `Ref. 1` · Quote found on the page · page could not be fetched; check by hand
- `Ref. 4` · Quote found on the page · not on the page, copy of 2026-11-03

U.S. Patent No. 7,479,949 B2

Touch screen device, method, and graphical user interface for determining commands by applying heuristics

CHARTED AGAINST iPhone multi-touch scrolling and swipe gestures, Apple

Patent owner's own product.

PATENT RECORD — EXHIBIT A-4

ASSIGNEE	Apple Inc
PRIORITY	Sept. 6, 2006
ISSUED	Jan. 20, 2009
EXPIRES	Sept. 5, 2027
STATUS	Active

From patents.google.com, copy of Nov. 3, 2026. 3 of 3 record checks passed.

12	● 8	◐ 4	○ 0	7/12	7	74/87	Nov. 3, 2026
CLAIM ELEMENTS	DOCUMENTED	PARTIAL	NOT FOUND	QUOTES VERBATIM ON PAGE	SOURCES QUOTED	CHECKS PASSED	RESEARCHED AND CHECKED

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[pre] A computing device, comprising:	● DOCUMENTED > "Both devices are powered by Apple A8 APL1011 system-on-chip processor and M8 second-generation motion coprocessor, with a Qualcomm MDM9625M LTE modem connecting to 4G."[1] The teardown report identifies computing hardware in iPhone 6 and iPhone 6 Plus, providing a concrete iPhone implementation for the device requirement.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · verbatim on an older copy of 2026-01-13; a fresh copy was not available ● Non-patent source ● In research basis ● Dated Dated Sept. 19, 2014, during the patent term.
1[a] a touch screen display;	● DOCUMENTED > "A few simple gestures—tap, drag, swipe, and pinch—are all you need to use iPhone and its apps."[2] The iPhone user guide describes touchscreen interaction through tapping, dragging, swiping and pinching. Apple's iPhone 6 technical specifications separately identify its display as Multi-Touch.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 15, 2015, during the patent term. 1 more supporting source in Appendix B.

1. *See inside the iPhone 6 and 6 Plus with new teardown - CNET*, cnet.com (Sept. 19, 2014), https://www.cnet.com/tech/mobile/see-inside-the-iphone-6-and-6-plus-with-new-teardown (last visited Jan. 13, 2026).
2. *iPhone User Guide*, cdsassets.apple.com (June 15, 2015), https://cdsassets.apple.com/live/6GJYWVAV/user/ma1724_iphone_user_guide.pdf (last visited Nov. 3, 2026).

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[b] one or more processors;	● DOCUMENTED > "Both devices are powered by Apple A8 APL1011 system-on-chip processor and M8 second-generation motion coprocessor, with a Qualcomm MDM9625M LTE modem connecting to 4G."[3] The report identifies the A8 application processor and M8 motion coprocessor in iPhone 6 and iPhone 6 Plus.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · verbatim on an older copy of 2026-01-13; a fresh copy was not available ● Non-patent source ● In research basis ● Dated Dated Sept. 19, 2014, during the patent term.
1[c] memory; and	● DOCUMENTED > "When an iOS device is turned on, its application processor immediately executes code from read-only memory known as the Boot ROM."[4] Apple documents read-only memory containing executable boot code in iOS devices.	CONFIDENCE: HIGH ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Oct. 2014, during the patent term.
1[d] one or more programs, wherein the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including:	● DOCUMENTED > "When an iOS device is turned on, its application processor immediately executes code from read-only memory known as the Boot ROM."[5] Apple explicitly describes stored program code being executed by the application processor. The guide also describes executable app memory pages being checked as they are loaded, supporting the general stored-program architecture rather than identifying individual gesture routines.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated Oct. 2014, during the patent term.
1[e] instructions for detecting one or more finger contacts with the touch screen display;	● DOCUMENTED > "In its default configuration, a view receives only the first UITouch object associated with an event, even if more than one finger is touching the view."[6] UIKit represents finger contacts as touch objects and delivers them to views for event handling. The documentation expressly addresses events involving one or more fingers.	CONFIDENCE: HIGH ● Quote and source ● Quote on page · 81% of the wording matches, copy of 2026-11-03 ● Non-patent source ● In research basis ● Dated · no date on the page

3. *See inside the iPhone 6 and 6 Plus with new teardown - CNET*, *supra* note 1.
4. *iOS Security*, apple.com (Oct. 2014), `https://www.apple.com/lae/privacy/docs/iOS_Security_Guide_Oct_2014.pdf` (last visited Nov. 3, 2026).
5. *Id.*
6. *Handling touches in your view | Apple Developer Documentation*, developer.apple.com, `https://developer.apple.com/documentation/uikit/handling-touches-in-your-view` (last visited Nov. 3, 2026).

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[f] instructions for applying one or more heuristics to the one or more finger contacts to determine a command for the device; and	PARTIAL > "Gesture recognizers encapsulate all of the logic needed to process and interpret incoming events for that view and match them to a known pattern."[7] UIKit documents pattern-based interpretation of touch events and notification of a target when a match occurs. This supports gesture-to-action interpretation, but does not disclose the particular heuristic rules used by built-in iPhone scrolling and swipe implementations.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page
1[g] instructions for processing the command;	DOCUMENTED > "Recognition of a gesture results in the dispatch of an action message to a target for each of the associated pairs."[8] UIKit dispatches an action message to the associated target after recognizing a gesture, providing the mechanism for processing the recognized user command.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 1 more supporting source in Appendix B.
1[h] wherein the one or more heuristics comprise:	PARTIAL > "If this property is true and the user begins dragging in one general direction (horizontally or vertically), the scroll view disables scrolling in the other direction. If the drag direction is diagonal, then scrolling doesn't lock and the user can drag in any direction until the drag completes."[9] Apple documents directional locking and unrestricted diagonal dragging; its iPhone guide separately documents swipe-based photo navigation. These sources do not establish that the three specified heuristics operate together as the claimed initial-angle-based selection scheme.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · 91% of the wording matches, copy of 2026-11-03 ● Non-patent source ● In research basis ● Dated · no date on the page 1 more supporting source in Appendix B.

7. *Handling UIKit gestures | Apple Developer Documentation*, developer.apple.com, `https://developer.apple.com/documentation/uikit/handling-uikit-gestures` (last visited Nov. 3, 2026).
8. *UIGestureRecognizer | Apple Developer Documentation*, developer.apple.com, `https://developer.apple.com/documentation/uikit/uigesturerecognizer?language=objc` (last visited Nov. 3, 2026).
9. *isDirectionalLockEnabled | Apple Developer Documentation*, developer.apple.com, `https://developer.apple.com/documentation/uikit/uiscrollview/isdirectionallockenabled` (last visited Nov. 3, 2026).

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[i] a vertical screen scrolling heuristic for determining that the one or more finger contacts correspond to a one-dimensional vertical screen scrolling command rather than a two-dimensional screen translation command based on an angle of initial movement of a finger contact with respect to the touch screen display;	◐ PARTIAL > "If this property is true and the user begins dragging in one general direction (horizontally or vertically), the scroll view disables scrolling in the other direction."[10] When directional locking is enabled, a generally vertical initial drag disables horizontal scrolling. The documentation does not specify an initial-movement-angle test or establish that this optional setting, whose default is false, is enabled in the relevant built-in iPhone views.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · 81% of the wording matches, copy of 2026-11-03 ● Non-patent source ● In research basis ● Dated · no date on the page
1[j] a two-dimensional screen translation heuristic for determining that the one or more finger contacts correspond to the two-dimensional screen translation command rather than the one-dimensional vertical screen scrolling command based on the angle of initial movement of the finger contact with respect to the touch screen display; and	◐ PARTIAL > "If the drag direction is diagonal, then scrolling doesn't lock and the user can drag in any direction until the drag completes."[11] Apple documents unrestricted dragging rather than axis locking when the drag is diagonal. It does not disclose how the initial movement angle is evaluated to select two-dimensional translation over vertical scrolling, or confirm the configuration of particular built-in iPhone views.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page
1[k] a next item heuristic for determining that the one or more finger contacts correspond to a command to transition from displaying a respective item in a set of items to displaying a next item in the set of items.	● DOCUMENTED > "While viewing a photo or video, tap to show and hide the controls. Swipe left or right to go forward or backward."[12] The Photos instructions map a left or right swipe to moving forward or backward through displayed photos or videos, documenting swipe-to-next-item behavior.	CONFIDENCE: HIGH ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated Dated June 15, 2015, during the patent term.

10. *Id.*
11. *Id.*
12. *iPhone User Guide*, *supra* note 2.

Appendix A · Prior art

Non-patent literature published before the priority date of Sept. 6, 2006. Dates are checked in Python; anything dated on or after it, or undated, is excluded.

REF.	PUBLISHED	REFERENCE	DISCLOSES	PASSAGE	CHECKS
1	2005-04	*AppLens and LaunchTile: Two Designs for One-Handed Thumb Use on Small Devices* Amy K. Karlson, Benjamin B. Bederson, John SanGiovanni[13]	1[pre], 1[a], 1[e], 1[f], 1[g]	"Each AppLens gesture is uniquely defined by a slope and direction, or vector, which allows gestures to be robust and highly personalizable; users can issue gestures of any length (beyond an activation threshold of 20 pixels) anywhere on the touch-sensitive surface."	● Before priority date ● Non-patent source ● Quote on page · 89% of the wording matches, copy of 2026-11-03 ● Labels match claim
2	2005-10	*Gesture Registration, Relaxation, and Reuse for Multi-Point Direct-Touch Surfaces* Mike Wu, Chia Shen, Kathy Ryall, Clifton Forlines, Ravin Balakrishnan[14]	1[pre], 1[a], 1[e], 1[f], 1[g]	"When any two fingers are placed onto the table (Figure 2a), the gesture is registered and the stylus behaves as a writing tool."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
3	2001	*Multi-Touch: A New Tactile 2-D Gesture Interface for Human-Computer Interaction* Wayne Westerman, John G. Elias, Alan Hedge[15]	1[f], 1[g]	"Asynchronous touches on a MTS are interpreted as conventional single-finger typing, while motions initiated by chords are interpreted as pointing, clicking, and other gesture commands."	● Before priority date ● Non-patent source ● Quote on page · 62% of the wording matches, copy of 2026-11-03 ● Labels match claim

13. Amy K. Karlson, Benjamin B. Bederson, John SanGiovanni, *AppLens and LaunchTile: Two Designs for One-Handed Thumb Use on Small Devices* (Apr. 2005), http://www.cs.umd.edu/hcil/jazz/learn/papers/chi-applens-launchtile.pdf (last visited Nov. 3, 2026).

14. Mike Wu, Chia Shen, Kathy Ryall, Clifton Forlines, Ravin Balakrishnan, *Gesture Registration, Relaxation, and Reuse for Multi-Point Direct-Touch Surfaces* (Oct. 2005), https://www.merl.com/publications/docs/TR2005-109.pdf (last visited Nov. 3, 2026).

15. Wayne Westerman, John G. Elias, Alan Hedge, *Multi-Touch: A New Tactile 2-D Gesture Interface for Human-Computer Interaction* (2001), https://www.openexhibits.org/wp-content/uploads/papers/HFES_multitouch_proceedings.pdf (last visited Nov. 3, 2026).

Appendix B · Research basis

For each element: Parallel's reasoning, its confidence, and every source it cited. The quoted source is marked ▸.

1[pre] DOCUMENTED · CONFIDENCE MEDIUM

The iPhone teardown report identifies the processors and connectivity hardware in the two iPhone models. The displayed publication date supplies the source date; the computing-device characterization follows from the identified hardware.

▸ See inside the iPhone 6 and 6 Plus with new teardown - CNET · www.cnet.com/tech/mobile/see-inside-the-iphone-6-and-6-plus-with-new-…

1[a] DOCUMENTED · CONFIDENCE MEDIUM

The user guide supplies the quoted gesture-operation sentence, and Apple's specifications explicitly identify the iPhone 6 Multi-Touch display. The mapping combines these two descriptions; no publication date is assigned to the undated guide.

▸ iPhone User Guide · cdsassets.apple.com/live/6GJYWVAV/user/ma1724_iphone_user_guide.pdf
iPhone 6 - Technical Specifications - Apple Support · support.apple.com/en-us/111954

1[b] DOCUMENTED · CONFIDENCE MEDIUM

The report explicitly identifies the A8 processor and M8 coprocessor. It is a secondary teardown report, so confidence is medium despite the direct statement.

▸ See inside the iPhone 6 and 6 Plus with new teardown - CNET · www.cnet.com/tech/mobile/see-inside-the-iphone-6-and-6-plus-with-new-…

1[c] DOCUMENTED · CONFIDENCE HIGH

Apple directly identifies read-only memory holding executable code. The guide's displayed October 2014 date supports the source date.

▸ iOS_Security_Guide_v39_FF · www.apple.com/lae/privacy/docs/iOS_Security_Guide_Oct_2014.pdf

1[d] DOCUMENTED · CONFIDENCE MEDIUM

Processor execution from ROM directly supports stored executable programs, while executable app memory pages support the application-software context. The source does not individually identify the gesture routines, so confidence in the overall mapping is medium.

▸ iOS_Security_Guide_v39_FF · www.apple.com/lae/privacy/docs/iOS_Security_Guide_Oct_2014.pdf

1[e] DOCUMENTED · CONFIDENCE HIGH

Apple documents touch objects delivered with events and expressly discusses multiple fingers touching a view. This directly supports software handling finger-contact events.

▸ Handling touches in your view | Apple Developer Documentation · developer.apple.com/documentation/uikit/handling-touches-in-your-view

1[f] PARTIAL · CONFIDENCE MEDIUM

Matching incoming events against a known pattern and notifying a target supports gesture interpretation leading to an action. It does not identify the particular heuristic implementation in stock iPhone scrolling and swiping, which is why the assessment is partial.

▸ Handling UIKit gestures | Apple Developer Documentation · developer.apple.com/documentation/uikit/handling-uikit-gestures

1[g] DOCUMENTED · CONFIDENCE MEDIUM

The recognizer dispatches an action message, and Apple's handling documentation explains that an action method provides a response. These describe processing a recognized gesture command, although they are framework-level rather than app-specific descriptions.

▸ UIGestureRecognizer | Apple Developer Documentation · developer.apple.com/documentation/uikit/uigesturerecognizer?language=objc
Handling UIKit gestures | Apple Developer Documentation · developer.apple.com/documentation/uikit/handling-uikit-gestures

1[h] PARTIAL · CONFIDENCE MEDIUM

The sources document directional locking, unrestricted diagonal dragging and photo swipe navigation separately. They do not establish a single combined implementation of the three specified heuristics or its initial-angle selection logic.

▸ isDirectionalLockEnabled | Apple Developer Documentation · developer.apple.com/documentation/uikit/uiscrollview/isdirectionalloc…
iPhone User Guide · cdsassets.apple.com/live/6GJYWVAV/user/ma1724_iphone_user_guide.pdf

1[i] PARTIAL · CONFIDENCE MEDIUM

The enabled property locks scrolling to the general direction in which dragging begins, including vertical dragging. Its default is false; the source does not disclose the angle test or demonstrate enabling it in a particular built-in view.

‣ isDirectionalLockEnabled | Apple Developer Documentation · developer.apple.com/documentation/uikit/uiscrollview/isdirectionalloc…

1[j] PARTIAL · CONFIDENCE MEDIUM

Diagonal dragging is expressly described as remaining unlocked in any direction, supporting two-dimensional movement. The exact initial-angle decision mechanism and app-specific use remain undocumented by this source.

‣ isDirectionalLockEnabled | Apple Developer Documentation · developer.apple.com/documentation/uikit/uiscrollview/isdirectionalloc…

1[k] DOCUMENTED · CONFIDENCE HIGH

The contiguous Photos instructions directly associate left/right swipes with forward/backward item navigation, supporting the functional next-item behavior. No internal recognition algorithm is inferred.

‣ iPhone User Guide · cdsassets.apple.com/live/6GJYWVAV/user/ma1724_iphone_user_guide.pdf

Appendix C · Verification log

Every check run on this exhibit. Green passed, amber needs a person to look, red failed; details for every flag follow the grid.

CLAIM LANGUAGE
Extract API on patents.google.com, copy of Nov. 3, 2026. Split into elements in Python; no model writes claim text.

EVIDENCE
Task API, pro processor, patent databases excluded.
`srun_728f2b03f51547d3aeca29547d158533`

PRIOR ART
Task API, pro processor, dates enforced in Python.
`srun_728f2b03f51547d3ba5fac1e6bb07b7c`

QUOTES, SOURCES AND DATES
Extract API re-reads every cited page from a copy at most a day old; Python compares each quote word for word and checks domains and dates.

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parallel

PATENT RECORD	PATENT NUMBER	CLAIM 1 VERBATIM	DATES IN ORDER
US7479949B2	●	●	●

CLAIM ELEMENT	RESEARCHED	QUOTE AND SOURCE	QUOTE ON PAGE	NON-PATENT SOURCE	IN RESEARCH BASIS	DATED
1[pre]	●	●	●	●	●	●
1[a]	●	●	●	●	●	●
1[b]	●	●	●	●	●	●
1[c]	●	●	●	●	●	●
1[d]	●	●	●	●	●	●
1[e]	●	●	●	●	●	●
1[f]	●	●	●	●	●	●
1[g]	●	●	●	●	●	●
1[h]	●	●	●	●	●	●
1[i]	●	●	●	●	●	●
1[j]	●	●	●	●	●	●
1[k]	●	●	●	●	●	●

PRIOR ART	BEFORE PRIORITY DATE	NON-PATENT SOURCE	QUOTE ON PAGE	LABELS MATCH CLAIM
Ref. 1	●	●	●	●
Ref. 2	●	●	●	●
Ref. 3	●	●	●	●

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- `1[pre]` · Quote found on the page · verbatim on an older copy of 2026-01-13; a fresh copy was not available
- `1[b]` · Quote found on the page · verbatim on an older copy of 2026-01-13; a fresh copy was not available
- `1[e]` · Quote found on the page · 81% of the wording matches, copy of 2026-11-03
- `1[e]` · Source dated · no date on the page
- `1[f]` · Source dated · no date on the page
- `1[g]` · Source dated · no date on the page
- `1[h]` · Quote found on the page · 91% of the wording matches, copy of 2026-11-03
- `1[h]` · Source dated · no date on the page
- `1[i]` · Quote found on the page · 81% of the wording matches, copy of 2026-11-03
- `1[i]` · Source dated · no date on the page
- `1[j]` · Source dated · no date on the page
- `Ref. 1` · Quote found on the page · 89% of the wording matches, copy of 2026-11-03
- `Ref. 3` · Quote found on the page · 62% of the wording matches, copy of 2026-11-03

U.S. Patent No. 6,285,999 B1

Method for node ranking in a linked database

CHARTED AGAINST Google Search ranking by links (PageRank), Google

Patent assigned to Google and Stanford; expired in 2018.

PATENT RECORD — EXHIBIT A-5

ASSIGNEE	Google LLC
PRIORITY	Jan. 10, 1997
ISSUED	Sept. 4, 2001
EXPIRED	Jan. 9, 2018
STATUS	Expired - Lifetime

From patents.google.com, copy of Oct. 29, 2026. 3 of 3 record checks passed.

4	● 4	◐ 0	○ 0	3/4	2	35/39	Nov. 3, 2026
CLAIM ELEMENTS	DOCUMENTED	PARTIAL	NOT FOUND	QUOTES VERBATIM ON PAGE	SOURCES QUOTED	CHECKS PASSED	RESEARCHED AND CHECKED

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[pre] A computer implemented method of scoring a plurality of linked documents, comprising:	● DOCUMENTED > "We use more than 200 signals, including our patented PageRank™ algorithm, to examine the entire link structure of the web and determine which pages are most important."[1] Google describes PageRank as an algorithm for determining page importance from the web's link structure. Its engineering account also describes computer computation of the PageRank graph across millions of pages.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page 2 more supporting sources in Appendix B.
1[a] obtaining a plurality of documents, at least some of the documents being linked documents, at least some of the documents being linking documents, and at least some of the documents being both linked documents and linking documents, each of the linked documents being pointed to by a link in one or more of the linking documents;	● DOCUMENTED > "We start at a set of well-connected initial pages and follow each of their links to new pages. Then we follow the links on those new pages to even more pages and so on, until we have a huge list of links."[2] Google describes discovering pages by following links and then following outgoing links on the newly discovered pages. Those intermediate pages are both linked-to documents and linking documents.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page · not on the page, copy of 2026-11-03 ● Non-patent source ● In research basis ● Dated Dated July 25, 2008, during the patent term.

1. *Technology Overview*, web.archive.org, https://web.archive.org/web/20100210175913/http:/www.google.com/corporate/tech.html (last visited Nov. 3, 2026).
2. *We knew the web was big...*, googleblog.blogspot.com (July 25, 2008), http://googleblog.blogspot.com/2008/07/we-knew-web-was-big.html (last visited Nov. 3, 2026).

CLAIM 1 (VERBATIM)	PUBLIC EVIDENCE	BASIS AND CHECKS
1[b] assigning a score to each of the linked documents based on scores of the one or more linking documents and	● DOCUMENTED > "PageRank also considers the importance of each page that casts a vote, as votes from some pages are considered to have greater value, thus giving the linked page greater value."[3] Google explains that a linking page's importance affects the value its link contributes to the linked page. This describes assigning linked-page importance based on the importance of linking pages.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page
1[c] processing the linked documents according to their scores.	● DOCUMENTED > "Pages that we believe are important pages receive a higher PageRank and are more likely to appear at the top of the search results."[4] Google describes using PageRank to influence the ordering of search results: higher-scored pages are more likely to appear near the top. PageRank is one ranking factor, not the sole determinant of result order.	CONFIDENCE: MEDIUM ● Quote and source ● Quote on page ● Non-patent source ● In research basis ● Dated · no date on the page

3. *Technology Overview*, *supra* note 1.
4. *Id.*

Appendix A · Prior art

Non-patent literature published before the priority date of Jan. 10, 1997. Dates are checked in Python; anything dated on or after it, or undated, is excluded.

REF.	PUBLISHED	REFERENCE	DISCLOSES	PASSAGE	CHECKS
1	1996	*Search and Ranking Algorithms for Locating Resources on the World Wide Web* Budi Yuwono; Dik L. Lee [5]	1[pre], 1[a], 1[b], 1[c]	"then the score of a document is propagated to the documents it references."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
2	1987-11	*Searching for Information in a Hypertext Medical Handbook* Mark Edwin Frisse [6]	1[pre], 1[a], 1[c]	"We used these syntactic relationships to propagate query-generated card weights through the hypertext document."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim
3	1989-08-10	*A Retrieval Model Incorporating Hypertext Links* W. B. Croft; H. Turtle [7]	1[pre], 1[a], 1[c]	"This constraint is applied to each N node in turn (conceptually, at least) and the end result is a ranking of hypertext nodes according to the values of P(Q)."	● Before priority date ● Non-patent source ● Quote on page ● Labels match claim

5. Budi Yuwono; Dik L. Lee, *Search and Ranking Algorithms for Locating Resources on the World Wide Web* (1996), https://www.cse.ust.hk/~dlee/Papers/ir/icde-wise-1996.pdf (last visited Nov. 3, 2026).
6. Mark Edwin Frisse, *Searching for Information in a Hypertext Medical Handbook* (Nov. 1987), https://dl.acm.org/doi/pdf/10.1145/317426.317433 (last visited Nov. 3, 2026).
7. W. B. Croft; H. Turtle, *A Retrieval Model Incorporating Hypertext Links* (Aug. 10, 1989), https://web.cs.umass.edu/publication/docs/1989/UM-CS-1989-084.pdf (last visited Nov. 3, 2026).

Appendix B · Research basis

For each element: Parallel's reasoning, its confidence, and every source it cited. The quoted source is marked ▸.

1[pre] DOCUMENTED · CONFIDENCE MEDIUM

The quoted corporate description identifies PageRank as an algorithm analyzing linked pages to determine importance. The engineering blog explicitly describes workstation computation over 26 million pages, while the architecture description states that the links database is used to compute PageRanks for all documents. The computer-implementation support includes a retrospective account of the early system, and the quoted archived overview is undated.

▸ Corporate Information - Technology Overview · web.archive.org/web/20100210175913/http:/www.google.com/corporate/tech.html
Official Google Blog: We knew the web was big... · http://googleblog.blogspot.com/2008/07/we-knew-web-was-big.html
The Anatomy of a Large-Scale Hypertextual Web Search Engine · www.cs.cmu.edu/~christos/courses/826-resources/PAPERS%2BBOOK/google-a…

1[a] DOCUMENTED · CONFIDENCE MEDIUM

The two consecutive sentences describe following links from initial pages to new pages and then following links on those new pages. The intermediate pages therefore have incoming links and outgoing links, supporting the linked-and-linking document roles. The stated publication date falls within the supplied patent term; the mapping from traversal to obtaining documents requires a modest synthesis.

▸ Official Google Blog: We knew the web was big... · http://googleblog.blogspot.com/2008/07/we-knew-web-was-big.html

1[b] DOCUMENTED · CONFIDENCE MEDIUM

The source explicitly states that the importance of a voting page affects the value given to the linked page. This directly supports importance-weighted contributions from linking documents. The archived source does not state a publication or update date.

▸ Corporate Information - Technology Overview · web.archive.org/web/20100210175913/http:/www.google.com/corporate/tech.html

1[c] DOCUMENTED · CONFIDENCE MEDIUM

The source directly connects higher PageRank with a greater likelihood of appearing at the top of search results. Its description of more than 200 signals supports the qualification that PageRank influences ordering without being the sole ranking factor. The archived source is undated.

▸ Corporate Information - Technology Overview · web.archive.org/web/20100210175913/http:/www.google.com/corporate/tech.html

Appendix C · Verification log

Every check run on this exhibit. Green passed, amber needs a person to look, red failed; details for every flag follow the grid.

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PRIOR ART
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QUOTES, SOURCES AND DATES
Extract API re-reads every cited page from a copy at most a day old; Python compares each quote word for word and checks domains and dates.

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parallel

PATENT RECORD	PATENT NUMBER	CLAIM 1 VERBATIM	DATES IN ORDER
US6285999B1	●	●	●

CLAIM ELEMENT	RESEARCHED	QUOTE AND SOURCE	QUOTE ON PAGE	NON-PATENT SOURCE	IN RESEARCH BASIS	DATED
1[pre]	●	●	●	●	●	●
1[a]	●	●	●	●	●	●
1[b]	●	●	●	●	●	●
1[c]	●	●	●	●	●	●

PRIOR ART	BEFORE PRIORITY DATE	NON-PATENT SOURCE	QUOTE ON PAGE	LABELS MATCH CLAIM
Ref. 1	●	●	●	●
Ref. 2	●	●	●	●
Ref. 3	●	●	●	●

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- ● 1[pre] · Source dated · no date on the page
- ● 1[a] · Quote found on the page · not on the page, copy of 2026-11-03
- ● 1[b] · Source dated · no date on the page
- ● 1[c] · Source dated · no date on the page